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UL 723 Surface Burning Characteristics of "Vision Building Systems" (90435)

A Report To:	**Vision Profile Extrusions Limited** 300 Zenway Blvd. Woodbridge, ON, Canada L4H 4M7
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Submitted by:	Element Fire Testing
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1.0 ACCREDITATION

ISO/IEC 17025 for a defined Scope of Testing by the American Association for Laboratory Accreditation (A2LA), Certificate Number: 6524.03.

2.0 SPECIFICATIONS OF ORDER

Determine the Flame Spread and Smoke Developed Indices based upon a single test conducted in accordance with UL 723 (Eleventh Edition), as per Vision Profile Extrusions Limited reference Purchase Order No. 22706 and Element Quotation No. 24-002-619054 dated November 1, 2024.

2.1 History of Revision

This report supersedes Element Test Report No. 24-002-431(B), originally issued on December 16, 2024. It is revised by request to update the company information and the sample description.

3.0 SAMPLE IDENTIFICATION

Material Identification	**"Vision Building Systems" (90435)**
Supplied Material Description*	Stay in Place PVC Formwork with Concrete (0.120" / 3.05mm PVC facing flame)
Material Thickness	2 inch (51 mm)
Date of Material Receipt	2024-12-12
Element Sample Identification No.	24-002-S0431-2
Test Date	2024-12-16

* The material description as supplied by the client. Element cannot always validate the accuracy of the description provided. If appreciable differences are observed between the material description and the specimen, test data may not be representative.

4.0 TEST PROCEDURE

The method, designated as UL 723 "*Test for Surface Burning Characteristics of Building Materials*" (procedurally consistent with ASTM E84), is designed to determine the relative surface burning under specific test conditions, where the material under test is mounted so that it forms the ceiling of a horizontal fire tunnel. A specified airflow is introduced through the tunnel and a specified flame is applied to one end. Observations are then made regarding the flame spread along the specimen surface. Results are expressed in terms of Calculated Flame Spread (CFS), Calculated Smoke Developed (CSD), Flame Spread Index (FSI) (CFS rounded to the nearest multiple of 5), and Smoke Developed Index (SDI) (CSD rounded accordingly).

Although the procedure is applicable to materials, products and assemblies used in building construction for development of comparative surface spread of flame data, the test results may not reflect the relative surface burning characteristics of tested materials under all building fire conditions.

5.0 SAMPLE PREPARATION

The test specimen consisted of a total of four sections of material, each approximately 21.25 inches (540 mm) in width by 72 inches (1829 mm) in length. The sections were butted together end-to-end to create the total specimen length. Prior to testing, the specimen was conditioned to constant weight at a temperature of 73 ± 5°F (23 ± 3°C) and a relative humidity of 50 ± 5%. At the time of test initiation, the specimen was self-supporting.

6.0 SUMMARY OF TEST PROCEDURE

The tunnel is preheated to 150 ± 5°F (66 ± 2.8°C), as measured by the floor-embedded thermocouple located 23.25 feet (7087 mm) downstream of the burner ports, and is allowed to cool to 105 ± 5°F (40.5 ± 2.8°C), as measured by the floor-embedded thermocouple located 13 feet (3962 mm) from the burners. The tunnel lid is then raised and the test specimen is placed along the ledges of the tunnel so as to form a continuous ceiling 24 feet (7315 mm) long, approximately 12 inches (305 mm) above the floor. Three 8 foot (2438 mm) sections of 0.25 inch (6 mm) cement board are then placed on the back side of the specimen and the lid is then lowered into place.

Upon ignition of the gas burners, the flame spread distance is observed and recorded every second. Flame spread distance versus time is plotted. Calculations ignore all flame front recessions and Flame Spread Index (FSI) is determined by calculating the total area under the curve for the test sample. If the area under the curve (A) is less than or equal to 97.5 min·ft, then FSI = 0.515·A; if greater, FSI = 4900/(195-A). FSI is then rounded to the nearest multiple of 5.

Smoke Developed Index (SDI) is determined by dividing the total area under the obscuration curve by that established for liquid heptane, and multiplying by 100. SDI is then rounded to the nearest multiple of 5 if less than 200. SDI values over 200 are rounded to the nearest multiple of 50.

7.0 TEST RESULTS

SAMPLE: "Vision Building Systems" (90435)

Approx. Time to Ignition (s)	Maximum Flame Front Distance	Time to Maximum Flame Front (s)	Maximum Temperature (° F)	Flame Spread Index (FSI)	Smoke Developed Index (SDI)
25	(ft.): 5.3 (m): 1.62	306	496	**25**	**400**

7.1 Observations of Burning Characteristics

The specimen ignited appoximately 27 seconds after exposure to the test burner flame. No melting or dripping behavior observed.

8.0 INTERPRETATION OF RESULTS

Industry documents such as the International Building Code (IBC), NFPA 101 Life Safety Code, etc. refer to UL 723 (ASTM E84, NFPA 255) test results using the following material classification categories:

	Flame-Spread Index (FSI)	Smoke Developed Index (SDI)
Class 1 or Class A	0 - 25	450 Maximum
Class 2 or Class B	26 - 75	450 Maximum
Class 3 or Class C	76 - 200	450 Maximum
Tested Material Results Classification (if applicable):		**Class 1 or Class A**





Francis Williams,
Technician.

Ian Smith,
Technical Manager.

Notes: This report is related only to the sample identified. It shall not be reproduced, except in full, without approval. It is covered under Element Materials Technology Canada Inc. Standard Terms and Conditions of Contract, which are accessible at www.element.com, or by calling 1-866-263-9268. UL 723 is a well-established test method that reports data in the form of indices. As such, MU cannot be calculated. In the reporting instructions, calculated values are rounded to the nearest multiple of 5 for FSI, and 5 or 50 for SDI, depending on the result. Since the rounding ranges establish precision and include potential uncertainty, by following the reporting instructions, the lab is considered to have satisfied the MU reporting requirements of ISO/IEC 17025.

9.0 TEST CHARTS **UL 723 - Eleventh Edition (ASTM E84-24)**

Sample: "Vision Building Systems" (90435)

Chart 1. FLAME SPREAD

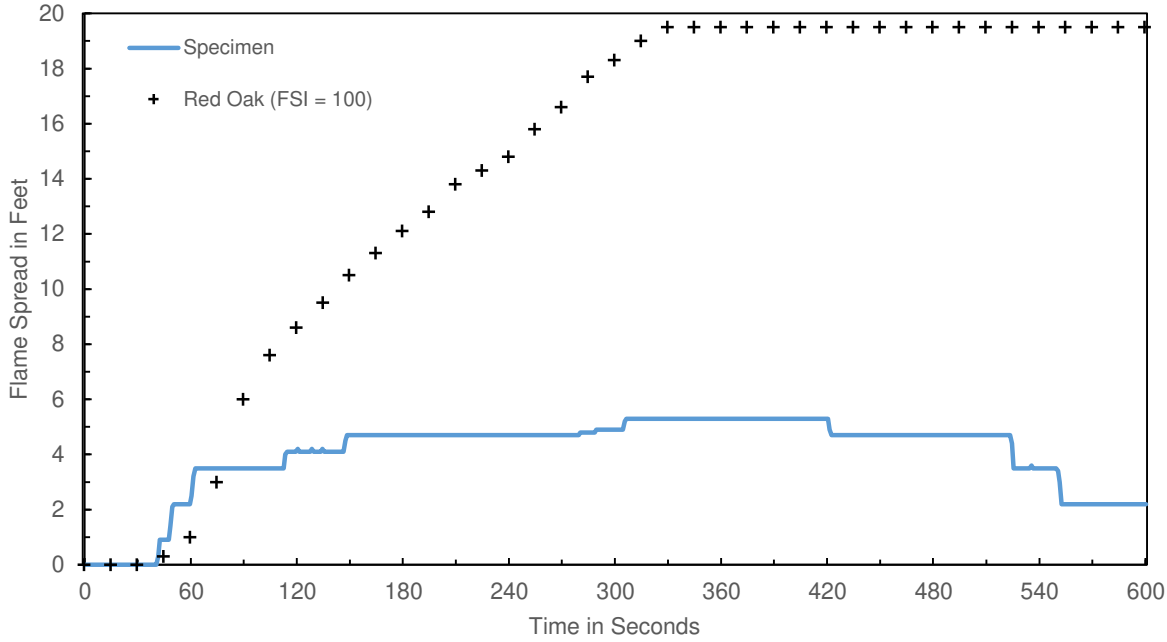
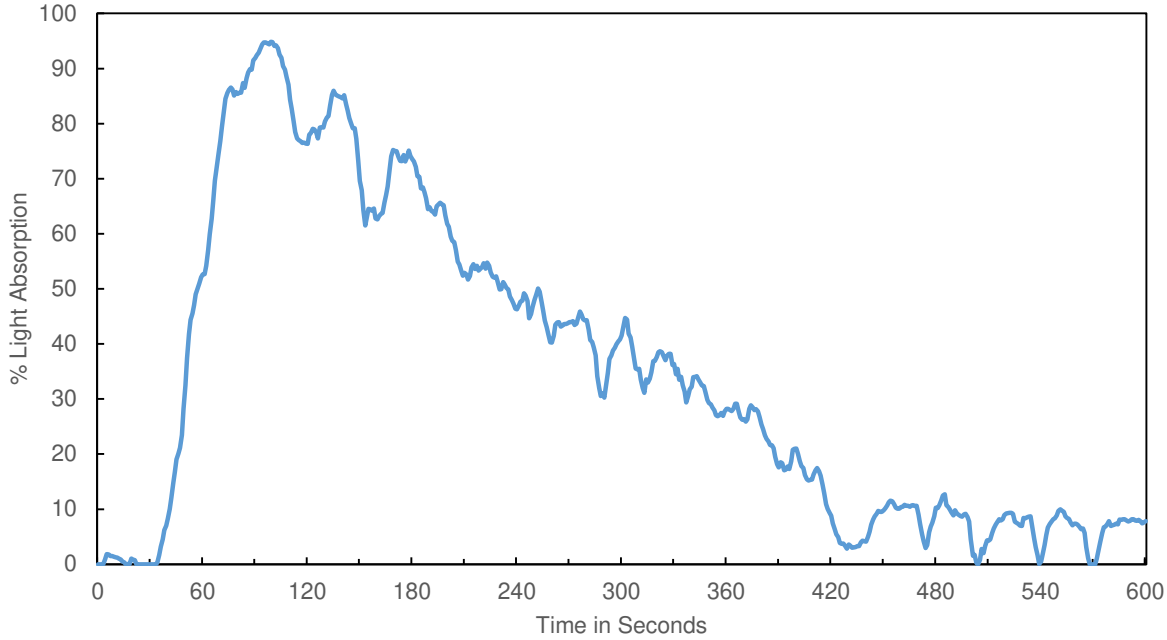


Chart 2. SMOKE DEVELOPED



Calculated Flame Spread (CFS)	Rounded Flame Spread Index (**FSI**)	Calculated Smoke Developed (CSD)	Rounded Smoke Developed Index (**SDI**)	Maximum 23' Air Temperature (°F)
22.9	**25**	382.4	**400**	496